

Advanced Lesson 3 - Exercises: Solve, Plot, Differentiate, Integrate, Simplify

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MATH 7013 - Advanced Mathematical Modeling

Everyday notation has been converted to Mathematica notation throughout: equations use "==", functions use square brackets, user definitions use the underscore, and E, Pi and I are the built-in constants rather than the bare letters e, p and i.

```
Clear[a, b, c, x, y, f, n];
```

1: Solving a biquadratic

```
Solve[a x^4 + b x^2 + c == 0, x]
```

```
Solve[a x^4 + b x^2 + c == 0, a]
```

Solving for x gives four roots, in the familiar nested-radical form: the quadratic formula applied to x^2 , then a square root taken of each branch.

Solving for a is trivial by comparison - the equation is linear in a, so there is exactly one solution, $a \rightarrow -(c + b x^2)/x^4$. Which variable you solve for changes the difficulty of the problem completely.

2: A nonlinear system in two unknowns

```
sol2a = Solve[{a x^2 + b y + c == z, y + a z^2 + b x^2 + c == 1}, {x, y}]
```

```
Simplify[sol2a]
```

Part 1. Solve returns two solution branches, each a pair of rules for x and y, and each carrying a square root. Simplify does not shorten this answer appreciably. Compare the two outputs: the radicand $b + c - b c - z - a b z^2$

is refactored into $c - z - b(-1 + c + a z^2)$ and the y component is returned unchanged. That is a rearrangement, not a reduction - the same number of symbols, differently bracketed. Simplify is not a magic shrink ray; on an answer that is already near its minimal form, its cost function finds nothing genuinely cheaper and it earns nothing but time.

```
sol2b = Solve[{a x^2 + b y + c == z, y + a z^2 + b x^2 + c == 1}, {a, x}]
```

Part 2. Solving the same system for a and x instead is a different problem and produces a different, generally messier, set of branches - a appears multiplied by x^2 in one equation and by z^2 in the other, so eliminating it does not decouple as cleanly.

3: Plot and integrate $E^{(2x)} \sin[3x] \cos[4x]$

```
Plot[E^2 x Sin[3 x] Cos[4 x], {x, 0, 2}]
```

```
int3 = Integrate[E^2 x Sin[3 x] Cos[4 x], x]
```

```
Simplify[int3]
```

The plot is an oscillation with an exponentially growing envelope: by $x = 2$ the amplitude is of order e^4 , so the wiggles near the origin are invisible at that vertical scale.

The antiderivative comes back already collected: a single factor of $E^{(2x)}$ multiplying a combination of $\cos[x]$, $\cos[7x]$, $\sin[x]$ and $\sin[7x]$ over 530. Note that the product of three oscillations has been rewritten as a sum of two, which is the product-to-sum identity applied twice.

Simplify returns that expression completely unchanged - the two output cells are identical. Integrate had already produced the collected form, so there was nothing left for Simplify to do. This is the opposite situation from Exercise 4, where the derivative arrives uncollected and Simplify does real work.

4: The same function, ten ways

```
Plot[E^2 x Sin[3 x] Cos[4 x], {x, 0, 2}]
```

```
d4 = D[E^2 x Sin[3 x] Cos[4 x], x]
```

```
Simplify[d4]
```

Parts 1 to 3. The derivative is a sum of three product terms by the product rule. Simplify collects the shared $E^{(2x)}$ and regroups the trigonometric factors, which does shorten it.

```

Clear[f, x];
f[x_] = e2 x Sin[3 x] Cos[4 x];
Plot[f[x], {x, 0, 2}]

f'[x]

Simplify[f'[x]]

{Simplify[f'[x] -  $\partial_x f[x]$ ]}

{Simplify[f''[x] -  $\partial_{\{x,2\}} f[x]$ ]}

{Simplify[f(3)[x] -  $\partial_{\{x,3\}} f[x]$ ]}

```

Parts 4 to 10. Immediate assignment is correct here: the right-hand side is a fixed expression in x with nothing to postpone, so $f[x_] = \dots$ stores the expression itself and every later call substitutes into it.

Each of the last three cells subtracts the two notations and simplifies. All three return zero, which is the verification the exercise asks for: the prime notation $f'[x]$ and the explicit $D[f[x], x]$ are the same operation written two ways, and the same holds for the second and third derivatives. Writing the difference and simplifying is a more honest check than eyeballing two large expressions side by side.

5: The local maximum of $x^3 - x + 1$

```

Clear[f, x];
f[x_] = x3 - x + 1;

{f'[x],  $\partial_x f[x]$ , Simplify[f'[x] -  $\partial_x f[x]$ ]}

{f''[x],  $\partial_{\{x,2\}} f[x]$ , Simplify[f''[x] -  $\partial_{\{x,2\}} f[x]$ ]}

Plot[{f[x], f'[x]}, {x, -2, 2}, PlotLegends → {"f", "f'"}]

crit = Solve[f'[x] == 0, x]

Table[{x /. s, f''[x] /. s, f[x] /. s}, {s, crit}]

```

Parts 1 to 3 confirm the two derivative notations agree, as in Exercise 4.

Part 4 plots the function against its derivative; the zeros of the derivative sit directly under the turning points of the function, which is the visual statement of the first-derivative test.

Part 5. The critical points are $x = -1/\sqrt{3}$ and $x = +1/\sqrt{3}$. The second derivative is $6x$, so at $x = -1/\sqrt{3}$ it is negative - concave down, a local maximum, with value $1 + 2/(3\sqrt{3})$. At $x = +1/\sqrt{3}$ it is positive - concave up, a local minimum. The maximum is local and not global: the cubic runs off to $+\infty$ as x grows, so no global maximum exists.

6: Simplifying Cos[n Pi] under assumptions

```
Clear[n];
Simplify[Cos[n Pi]]
Simplify[Cos[n Pi], n ∈ ℤ]
Simplify[Cos[n Pi],  $\frac{n}{2} \in \mathbb{Z}$ ]
Simplify[Cos[n Pi],  $\frac{n-1}{2} \in \mathbb{Z}$ ]
```

With no assumption there is nothing to do: n could be anything, so Cos[n Pi] comes back unchanged.

Assuming n is an integer, the answer is $(-1)^n$ - the cosine lands on +1 or -1 alternately, and Mathematica can say so without knowing which.

Assuming n is even, meaning n/2 is an integer, the answer collapses to 1.

Assuming n is odd, expressed as (n-1)/2 being an integer, the answer collapses to -1.

The pattern is the point: Simplify can only simplify as far as the assumptions permit, and supplying the right domain is the difference between an unchanged expression and a single number.

7: Plot and integrate x^5 Sin[x]

```
Plot[x^5 Sin[x], {x, -1, 1}]
```

$$\int x^5 \sin[x] \, dx$$

$$\int_2^3 x^5 \sin[x] \, dx$$

The integrand is even (odd times odd), so the plot is symmetric about the vertical axis and non-negative on [-1, 1].

The indefinite integral is the generic antiderivative: five applications of integration by parts, producing a polynomial in x times Cos[x] plus another times Sin[x], with no constant of integration attached.

The definite integral from 2 to 3 is returned as an exact combination of Sin[2], Cos[2], Sin[3] and Cos[3] with integer coefficients. Exact, not numeric - apply N if a number is wanted.

8: Sqrt[x] and the trouble with negative limits

```
Plot[Sqrt[x], {x, -1, 1}, PlotStyle -> Thickness[0.01]]
```

Part 1. Nothing is drawn for $-1 \leq x < 0$. Sqrt of a negative real is imaginary, Plot only draws real values, so the curve simply begins at the origin. Mathematica does not complain; it silently omits the points it cannot place on a real axis.

$$\int \sqrt{x} \, dx$$

$$\int_1^2 \sqrt{x} \, dx$$

$$\int_a^b \sqrt{x} \, dx$$

Part 2 gives the generic antiderivative $(2/3) x^{(3/2)}$. Part 3 gives the exact value $(4 \sqrt{2} - 2)/3$.

Part 4 gives $(2/3)(b^{(3/2)} - a^{(3/2)})$, but wrapped in a ConditionalExpression carrying a long side condition on a and b. With symbolic limits Mathematica cannot know whether the path from a to b crosses the branch cut of the square root, so it states the value together with the conditions under which that value is the right one. The condition is unreadable and that is fine - it is machine bookkeeping, and supplying an assumption, as Parts 5 and 6 do, makes it collapse.

It is not surprising that the answer depends on both a and b: a definite integral is the net signed area between the two limits, and moving either limit moves that area. There is nothing special about this integrand - every definite integral with symbolic limits does the same.

$$\text{Integrate}[\sqrt{x}, \{x, a, b\}, \text{Assumptions} \rightarrow 0 < a < b]$$

Part 5. Assuming $0 < a < b$, the answer is real and finite. Not surprising: the whole interval of integration lies where $\text{Sqrt}[x]$ is real-valued, continuous and bounded, so the integral is an ordinary area under an ordinary curve.

$$\text{Integrate}[\sqrt{x}, \{x, a, b\}, \text{Assumptions} \rightarrow a < 0 < b]$$

Part 6. Assuming $a < 0 < b$, the answer is still finite but no longer real: it carries a factor of i . Not surprising either. The integrand is imaginary on the whole stretch from a to 0, so the integral picks up an imaginary contribution from that stretch. It converges because $x^{(1/2)}$ has an integrable singularity - the exponent $1/2$ is greater than -1 - so the integral is perfectly finite. It is simply finite in the complex plane rather than on the real line.